

Midterm Report: Multi-Future Pedestrian Trajectory Forecasting with Intent-Trajectory Consistency

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Abstract

This report summarizes our midterm progress on pedestrian intention estimation and multi-future trajectory forecasting. Starting from a deterministic baseline inspired by PIEPredict, we build a preliminary multi-hypothesis predictor that outputs K candidate futures and evaluates them with set-based metrics such as minADE/minFDE. At the midterm stage, we focus on a controlled prototype setting and report initial quantitative and qualitative results, along with failure cases and the next-step plan toward intent-trajectory consistency.

1 Introduction

Pedestrian future motion is inherently multi-modal: under the same observed history, a pedestrian may continue straight, slow down, stop, or begin crossing. This ambiguity makes deterministic regression unsuitable in many scenes. Our project studies joint pedestrian intention estimation and future trajectory forecasting, with a focus on explicitly modeling one-to-many uncertainty via multiple future hypotheses.

At this stage, our goal is to establish a working baseline and a preliminary multi-future prototype, then evaluate whether increasing the number of hypotheses improves set-based trajectory metrics and qualitative coverage. This is basically corresponding with the progress from week1-week4 in the project proposal.

2 Problem Setup

We consider two coupled prediction tasks: the first is intention estimation, we will predict whether a pedestrian will cross in the near future. The second is trajectory forecasting, we will predict the future pedestrian motion over a fixed horizon.

Inputs may include past bounding boxes, center trajectories, and optional context signals. Outputs include an intention probability and one or more candidate future trajectories.

3 Method

3.1 Baseline

Our baseline follows the deterministic prediction setting: given past observations, the model outputs a single future trajectory. This baseline serves as a reference point for evaluating whether multi-future prediction improves coverage in ambiguous scenes.

3.2 Multi-Future Prediction

We extend the deterministic formulation to output $K(= 1, 5, 10)$ future hypotheses: $\{\hat{\mathbf{y}}^{(k)}\}_{k=1}^K$, where in the case $K = 1$ it degenerates to the baseline. A standard Best-of- K loss is

$$\mathcal{L}_{\text{BoK}} = \min_{k \in \{1, \dots, K\}} \frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{y}}_t^{(k)} - \mathbf{y}_t\|_2^2.$$

This objective encourages at least one hypothesis to match the realized future well.

3.3 Intent-Trajectory Consistency (Planned)

Our final project direction also includes an intent-trajectory consistency regularizer. At midterm, this component is still under development. The high-level idea is to align the predicted crossing probability with the probability mass assigned to crossing-like trajectory modes.

4 Experimental Setup

4.1 Dataset and Split

In the initial repository [2], it should have 6 sets of data (from set01 to set06 [3]), but currently, we only choose the largest set03 which contains half of the data to make sure our pipeline works. There are 19 videos in total in this set, and we split 01-14 to training set, 15-17 to validation set, and 18-19 to test set.

4.2 Motion Representation

Our current prototype uses only pedestrian bounding-box centers as the motion representation. For each frame, we convert the annotated bounding box into a 2D center coordinate and use the past 15 centers to predict the next 45 future centers. We normalize each trajectory relative to the last observed point in the history window. At the midterm stage, we have image crops but not consider appearance features, speed signals, or other contextual inputs; the goal is to evaluate a minimal multi-future trajectory predictor in a controlled setting.

4.3 Metrics Reported

- **Intention:** Minimum Training Loss.

- **Single-mode trajectory: ADE@1 and FDE@1.** Here, ADE@1 measures the average displacement error over all future time steps for the first predicted trajectory, while FDE@1 measures the final displacement error at the last predicted time step for that same trajectory. These two metrics reflect the quality of the model’s primary single-mode prediction.
- **Minimum from Multiple trajectories: minADE@K and minFDE@K.** For each test sample, we first compute the ADE and FDE for each of the K predicted futures, then take the minimum over the K candidates. Therefore, minADE@K and minFDE@K evaluate whether the model’s set of predicted trajectories contains at least one future that is close to the ground-truth trajectory, and they are especially useful for measuring coverage in multi-modal prediction settings.

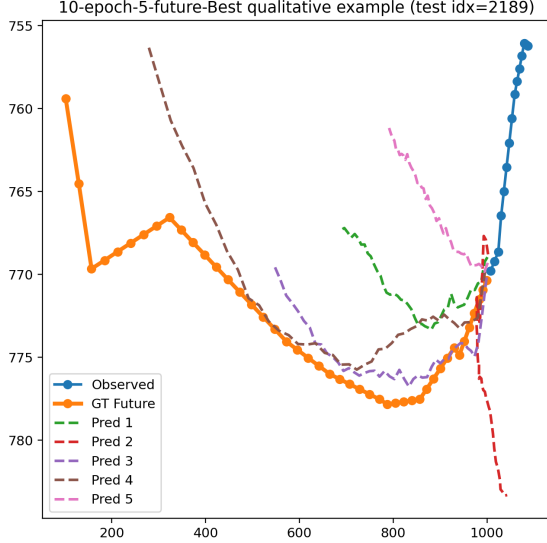
5 Midterm Results

5.1 Baseline and Multi-Future Quantitative Results

Table 1: Best Performance across different Ks and epochs.

K	Epoch	Loss	ADE@1	FDE@1	minADE@K	minFDE@K
1	10	1137.46	13.27	34.68	13.27	34.68
1	20	1101.57	13.27	34.68	13.27	34.68
1	30	1070.81	13.27	34.68	13.27	34.68
5	10	256.23	15.28	40.19	7.92	17.30
5	20	221.20	14.25	37.72	7.22	16.43
5	30	205.39	15.11	39.48	7.17	16.30
10	10	156.75	66.55	203.74	6.56	13.18
10	20	139.11	60.32	202.90	6.40	13.05

5.2 Qualitative Results Examples



(a) White-background trajectory visualization.



(b) Overlay on video frame / cropped scene visualization.

6 Discussion

6.1 What We Observe So Far

In section 5.1, we can see an obvious improvement on minADE@K, minFDE@K and training loss as we increasing K, and when we increase K from 1 to 5, the improvement is much larger than increasing K from 5 to 10. Moreover, it doesn't show performance improvement for ADE@1 and FDE@1 when we increase K from 1 to 5 and it becomes even much worse when we increase K from 5 to 10.

In section 5.2, we show two figures present the best example for $K = 5$ and epochs= 10, while the first one is the trajectory visualization on the white background, and the second one is a cropped frame/scene overlaid by the trajectories.

7 Limitations

The current midterm system is still a preliminary prototype and has several important limitations. At first, Our experiments are currently restricted to PIE set03 with a custom split, rather than the full official evaluation setting. The model uses only pedestrian bounding-box centers and does not yet incorporate image, context, or vehicle-speed information and the proposed intent-trajectory consistency component is not yet implemented in the current experiments. Moreover, improvements in minADE/minFDE mainly indicate better set coverage, and do not necessarily imply stronger single-mode prediction quality.

8 Next Steps

In the next stage, we plan to move beyond the current center-only prototype and integrate the multi-future formulation more fully into the PIEPredict-style pipeline, while adding an explicit intent-trajectory consistency component. We also aim to incorporate richer motion and context features, expand the experiments beyond the current restricted setting, and improve the qualitative analysis so that the predicted futures can be evaluated more clearly in realistic driving scenes.

References

- [1] Amir Rasouli, Iuliia Kotseruba, Toni Kunic, and John K. Tsotsos. PIE: A Large-Scale Dataset and Models for Pedestrian Intention Estimation and Trajectory Prediction. In *ICCV*, 2019.
- [2] PIEPredict repository. <https://github.com/aras62/PIEPredict>
- [3] PIEPredict full dataset. <https://drive.google.com/drive/folders/180MXX1z3aicZMwYu2pCM0TamzUKTOL16?dmr=1&ec=wgc-drive-%5Bmodule%5D-goto>